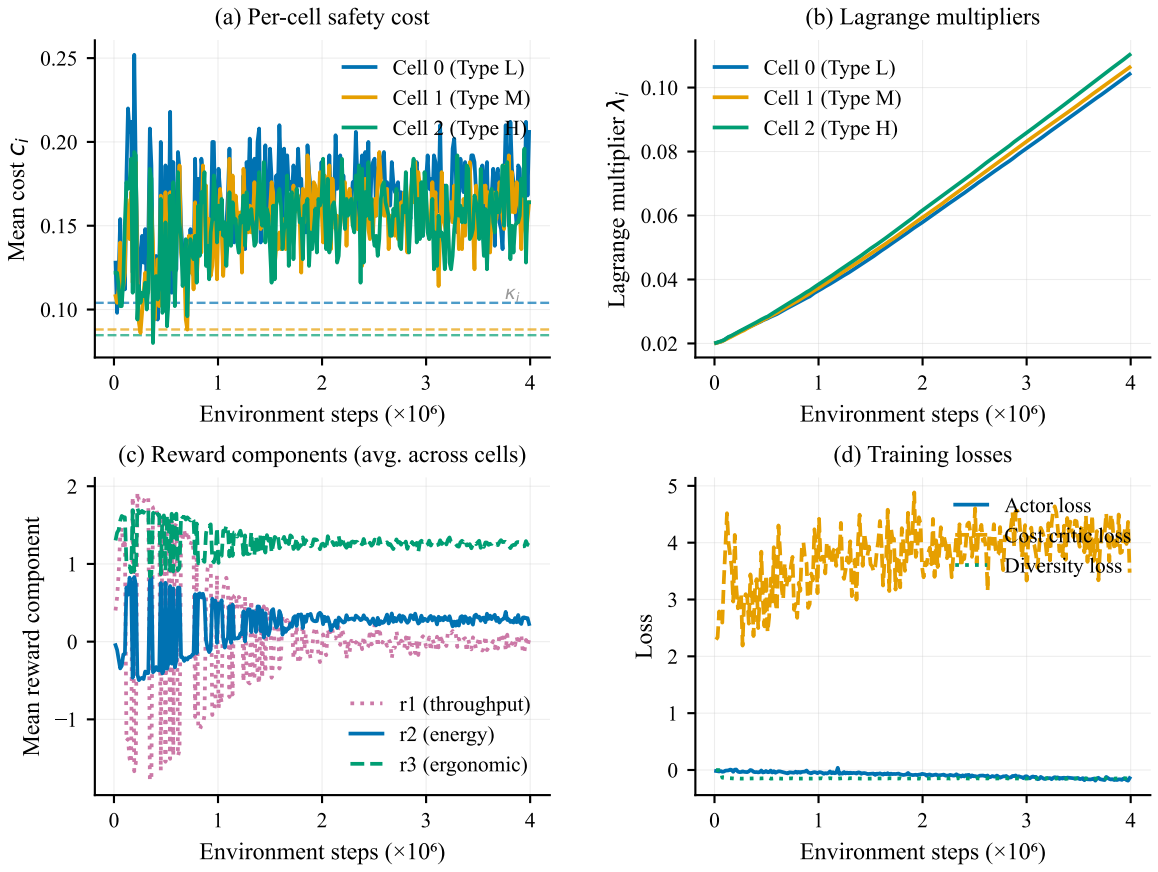
(a) Per-cell safety cost
Mean cost c_i
0.25
0.20
0.15
0.10
Cell 0 (Type L)
Cell 1 (Type M)
Cell 2 (Type H)
K_i
0
1
2
3
4
Environment steps (×10⁶)
(b) Lagrange multipliers
Lagrange multiplier λ_i
0.10
0.08
0.06
0.04
0.02
Cell 0 (Type L)
Cell 1 (Type M)
Cell 2 (Type H)
0
1
2
3
4
Environment steps (×10⁶)
(c) Reward components (avg. across cells)
Mean reward component
2
1
0
−1
r1 (throughput)
r2 (energy)
r3 (ergonomic)
0
1
2
3
4
Environment steps (×10⁶)
(d) Training losses
Loss
5
4
3
2
1
0
Actor loss
Cost critic loss
Diversity loss
0
1
2
3
4
Environment steps (×10⁶)